

AI Data Quality Agent

Problem Statement

Organizations often work with CSV datasets that contain missing values, duplicate records, and outliers. Poor-quality data leads to inaccurate analysis and poor business decisions.

The objective of this project is to build an AI-powered Data Quality Agent that automatically analyses a dataset, evaluates its quality, and generates meaningful business insights using Generative AI.

Objective

To develop a web application that:

- Uploads any CSV dataset.
- Detects missing values, duplicate records, and outliers.
- Calculates an overall Data Quality Score.
- Uses Gemini AI to generate business insights, data cleaning recommendations, and suggested dashboard KPIs.

Technologies Used

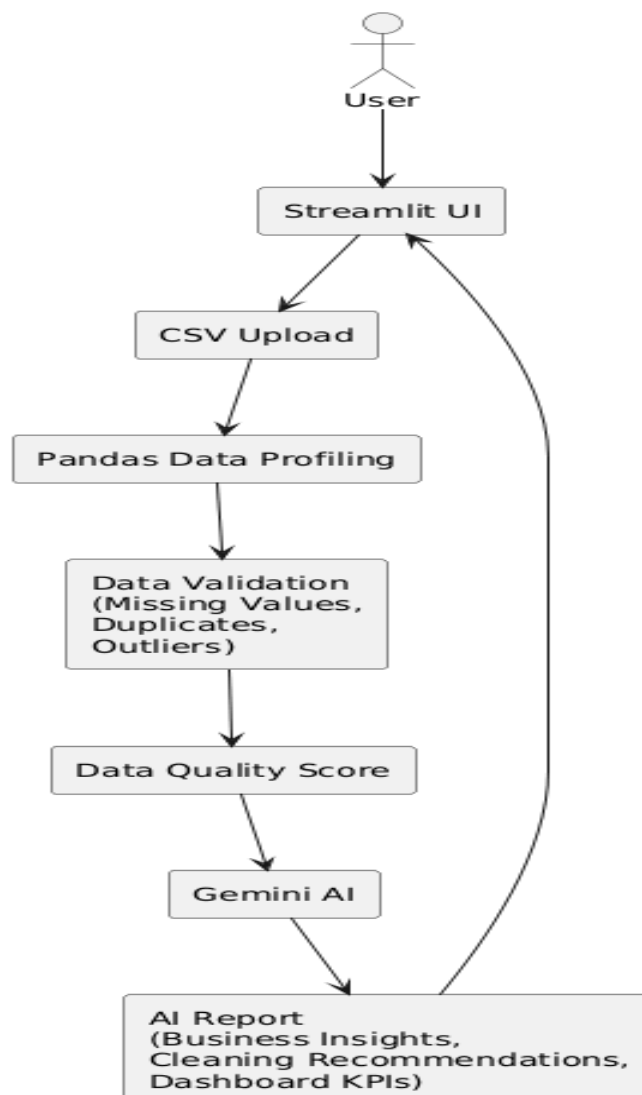
- **Python** – Core programming language.
- **Streamlit** – To build the interactive web application.
- **Pandas** – For reading and analyzing CSV datasets.
- **Google Gemini AI** – To generate business insights and recommendations.
- **Python Dotenv** – To securely manage the API key.

Project Workflow

1. User uploads a CSV dataset.
2. The dataset is read using Pandas.
3. The application performs data profiling:
 - Dataset Summary

- Missing Value Detection
 - Duplicate Detection
 - Outlier Detection (IQR Method)
4. A Data Quality Score is calculated based on the identified issues.
 5. The dataset summary is sent to Gemini AI.
 6. Gemini AI generates:
 - Business Insights
 - Data Cleaning Recommendations
 - Suggested Dashboard KPIs
 7. The complete report is displayed through the Streamlit interface.

System Architecture



Output

The application generates:

- Dataset Summary
- Missing Values Report
- Duplicate Records Report
- Outlier Detection Report
- Data Quality Score
- Overall Dataset Status
- AI-generated Business Insights
- Data Cleaning Recommendations
- Suggested Dashboard KPIs

Challenges Faced

- Integrating the Gemini API with the Streamlit application.
- Calculating a meaningful data quality score.
- Handling large datasets efficiently.
- Designing a simple and user-friendly interface within the hackathon time limit.

Future Enhancements

- Add interactive charts and visualizations.
- Export reports as PDF.
- Support Excel and JSON files.

Final Outcome

The AI Data Quality Agent successfully analyzes uploaded CSV datasets, evaluates their quality by detecting missing values, duplicates, and outliers, calculates a Data Quality Score, and leverages Gemini AI to generate business insights, cleaning recommendations, and dashboard KPIs.